

INTERNSHIP PROJECT REPORT

NIFTY 100 FINANCIAL INTELLIGENCE PLATFORM

A Financial Analytics, Screening and Intelligence Platform

Submitted by

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ABSTRACT

The Nifty 100 Financial Intelligence Platform is an integrated financial analytics system designed to convert structured financial and supporting datasets into meaningful company-level and market-level intelligence. The project addresses the practical challenge of analysing a large company universe consistently when financial information is distributed across multiple spreadsheets and data sources.

The platform performs data ingestion, cleaning, validation and SQLite storage; calculates financial ratios and growth indicators; provides configurable screening; performs peer and sector comparisons; analyses cash-flow behaviour; and generates reports and exports. A Streamlit dashboard and FastAPI service provide user-facing and programmatic access.

The project demonstrates how Python-based data engineering, relational database management, financial analytics, visualisation, API development and software testing can be combined into a practical financial-intelligence application.

Keywords: Financial Analytics, Nifty 100, Python, SQLite, Financial Ratios, Investment Screening, Peer Analysis, Streamlit, FastAPI, Data Engineering.

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1. INTRODUCTION

Financial analysis involves collecting information from income statements, balance sheets, cash-flow statements, market data and qualitative sources. When the number of companies increases, manual analysis becomes time-consuming and can introduce inconsistencies in calculations, naming conventions and comparison methods.

The Nifty 100 Financial Intelligence Platform was developed to provide a single analytical environment for the supplied Nifty 100 company universe. Instead of treating each spreadsheet as an isolated source, the platform establishes a common data model and connects datasets through company identifiers.

1.1 Need for the Project

A centralised system reduces repetitive work and ensures that the same definitions are applied consistently across the company universe.

1.2 Project Approach

- Profile the supplied datasets.
- Standardise identifiers and financial years.
- Validate and store the data.
- Calculate KPIs and intelligence.
- Expose outputs through dashboard, API and reports.

2. BACKGROUND AND PROBLEM STATEMENT

The source data is distributed across several Excel workbooks, each representing a different business or analytical perspective. Without an integration layer, users would repeatedly locate records, clean company names, align financial years and calculate ratios manually.

Problem statement: develop a self-contained financial intelligence platform capable of integrating the supplied datasets, calculating a broad set of financial indicators, supporting configurable screening and comparison, generating reports and providing interactive application access.

2.1 Challenges Addressed

- Heterogeneous spreadsheet structures.
- Inconsistent ticker and financial-year representations.
- Missing or incomplete observations.
- Duplicate and orphan records.
- Need for reproducible calculations.
- Need for peer and sector comparisons.

3. PROJECT OVERVIEW

The platform is organised as a pipeline from source data to user-facing intelligence. The design separates data preparation from analytics and separates analytics from presentation. This makes the analytical layer reusable across the dashboard, API and reports.

Layer	Responsibility	Output
Source	Original Excel datasets	Raw files
ETL	Cleaning and validation	Clean records, audit logs
Database	Structured storage	nifty100.db
Analytics	Financial calculations	Ratios, CAGR, scores
Intelligence	Screening and comparison	Screens, peer outputs
Reporting	Exports and reports	PDF, XLSX, CSV
Application	Interactive access	Streamlit
API	Programmatic access	FastAPI

The supplied company master contains 92 companies. This master acts as the reference universe for company-level analytics. Supporting records that cannot be linked to the universe are handled through validation rather than silently reassigned.

4. OBJECTIVES

The project objectives are aligned with the internship specification and the practical needs of a financial research workflow.

1. Build an automated financial data-ingestion and validation workflow.
2. Create a reliable SQLite database.
3. Calculate financial, growth, leverage and cash-flow KPIs.
4. Implement configurable investment screening.
5. Create financial-health analysis.
6. Provide peer-group comparison.
7. Provide sector-level analytics.
8. Analyse cash-flow behaviour.
9. Generate company and portfolio reports.
10. Develop an interactive Streamlit dashboard.
11. Expose selected analytics through FastAPI.
12. Create automated tests and data-quality checks.

Success is measured by the ability of the system to ingest the supplied data, produce consistent outputs, serve the dashboard, generate reports and pass the available automated validation suite.

5. REQUIREMENTS AND SCOPE

The project specification defines twelve major modules: data engineering and ETL, financial ratio engine, investment screener, financial health scoring, peer comparison, sector analytics, cash-flow intelligence, trend analysis, additional analytics, reporting/export, document intelligence and alerts/watchlist functionality.

5.1 In-Scope

- Financial statement integration.
- Financial ratio computation.
- Historical trend and CAGR analysis.
- Configurable screening.
- Financial health.
- Peer comparison.
- Sector analytics.
- Cash-flow intelligence.
- Reporting.
- Dashboard and API.

5.2 Out-of-Scope

- Real-time market feeds.
- Technical analysis as a primary module.
- Price prediction.
- Options and futures.
- Portfolio optimisation.
- ESG and sentiment-driven investment decisions.

6. DATASET DESCRIPTION

The project uses twelve supplied Excel datasets covering company master information, financial statements, analysis, documents, qualitative observations, sectors, market prices, market capitalisation, ratios and peer groups.

Dataset	Role
companies.xlsx	Company master and identifiers
profitandloss.xlsx	P&L and profitability
balancesheet.xlsx	Assets, liabilities, equity and debt
cashflow.xlsx	Cash-flow statements
analysis.xlsx	Growth and analysis
documents.xlsx	Annual reports and documents
prosandcons.xlsx	Qualitative observations
sectors.xlsx	Sector classification
stock_prices.xlsx	Historical market prices
market_cap.xlsx	Market-cap/valuation context
financial_ratios.xlsx	Reference ratios
peer_groups.xlsx	Peer relationships

Data preparation includes type conversion, ticker normalisation, financial-year normalisation, duplicate detection, missing-value inspection and relationship validation.

7. DATA ENGINEERING AND ETL

The ETL pipeline is the foundation of the platform. It converts heterogeneous Excel sources into a consistent relational representation.

Extraction reads the source workbooks. Transformation standardises identifiers, financial years and numeric values, while detecting duplicates and invalid relationships. Loading stores accepted records in SQLite.

The pipeline records processing outcomes in audit and validation outputs. This makes data-quality problems visible instead of silently discarding them.

7.1 Transformation Activities

- Trim whitespace and normalise tickers.
- Standardise financial-year representations.
- Convert numeric columns.
- Apply missing-value rules.
- Detect duplicates.
- Validate expected columns and relationships.

8. SYSTEM ARCHITECTURE

The architecture uses separate layers for source data, ETL, database storage, analytics, intelligence, reporting and application interfaces. This separation reduces coupling and makes the project easier to test and maintain.

Layer	Tools	Purpose
Data Source	Excel	Source information
ETL	Python, pandas	Cleaning and validation
Storage	SQLite	Persistent data
Analytics	pandas, NumPy, SciPy	KPI calculations
Intelligence	scikit-learn, rules	Clustering and derived intelligence
Visualisation	Plotly, Matplotlib	Charts
Reporting	ReportLab, openpyxl	Reports/exports
Application	Streamlit, FastAPI	Dashboard/API

Processing flow: Excel files → extraction → normalisation → validation → SQLite → KPI engine → screening/peer/sector/cash-flow analytics → reports/API/dashboard.

9. DATABASE DESIGN

SQLite is used as the central relational database because it is lightweight, portable and sufficient for the project workload. It keeps the application self-contained and avoids dependence on a separate database server.

9.1 Logical Entities

- Company master.
- Profit and loss.
- Balance sheet.
- Cash flow.
- Financial ratios.
- Market information.
- Sector classification.
- Peer groups.
- Documents.
- Qualitative observations.
- Derived analytics.

9.2 Integrity

Company identifiers provide the principal relationship across company-level datasets. Financial-year information aligns historical observations. Foreign-key and relationship checks identify orphan records.

10. FINANCIAL KPI METHODOLOGY

The KPI engine converts raw financial statement values into comparable analytical measures. The same definitions are reused by downstream modules to maintain consistency.

KPI	Formula / Interpretation
NPM	Net Profit / Sales × 100
OPM	Operating Profit / Sales × 100
ROE	Net Profit / Equity × 100
ROCE	EBIT / Capital Employed × 100
D/E	Debt / Equity
Interest Coverage	Operating profit and other income relative to interest
FCF	CFO + CFI
Asset Turnover	Sales / Total Assets
Revenue CAGR	Compound annual growth
PAT CAGR	Compound annual growth
EPS CAGR	Compound annual growth

Special handling is applied to zero denominators, negative values and turnaround cases. Where conventional CAGR is not meaningful, the system flags the condition rather than generating a misleading number.

11. INVESTMENT SCREENER

The investment screener converts financial conditions into repeatable filters. Thresholds can be configured rather than permanently embedded in the interface.

Screen	Illustrative Conditions
Quality Compounder	ROE > 15%; D/E < 1; FCF > 0; Revenue CAGR > 10%
Value Pick	P/E < 20; P/B < 3; D/E < 2; Dividend Yield > 1%
Growth Accelerator	PAT CAGR > 20%; Revenue CAGR > 15%; D/E < 2
Dividend Champion	Dividend Yield > 2%; Payout < 80%; FCF > 0
Debt-Free Blue Chip	D/E = 0; ROE > 12%; Revenue > ₹5,000 Cr
Turnaround Watch	Revenue CAGR > 10%; improving FCF; declining D/E

The output identifies qualifying companies and exposes the metrics used in the decision so that an analyst can review the screening logic.

12. FINANCIAL HEALTH ANALYSIS

The financial-health module provides a composite analytical view using four dimensions defined by the project specification.

Dimension	Weight	Examples
Profitability	35%	ROE, ROCE, NPM
Cash Quality	30%	FCF, CFO/PAT, FCF trend
Growth	20%	Revenue CAGR, PAT CAGR
Leverage	15%	D/E, Interest Coverage

The weighted structure balances profitability with cash quality, growth and financial risk. The resulting score is an analytical aid and should not be treated as an investment recommendation by itself.

13. PEER COMPARISON

Peer analysis compares companies with businesses in similar peer groups. This provides context that may be lost when a company is compared with the entire market.

13.1 Peer Metrics

- Profitability.
- Growth.
- Leverage.
- Cash flow.
- Valuation.
- Efficiency.

13.2 Percentile Analysis

Metrics are converted into relative positions within the peer group. This identifies companies that are above or below their peer distribution.

13.3 Radar Charts

Radar charts present multiple metrics together. The project generates 92 company-oriented radar outputs.

14. SECTOR ANALYTICS

Sector analytics aggregates company-level indicators according to the supplied sector classification. It helps determine whether a company's performance is typical or unusual within its sector.

14.1 Measures

- Sector median KPIs.
- Growth comparison.
- Leverage comparison.
- Revenue versus ROE.
- Market-cap context.
- Company-versus-sector comparison.

14.2 Data Note

The supplied sector workbook contains 10 broad sectors while the specification describes 11. The implementation uses the supplied classification and does not invent unsupported source data.

15. CASH-FLOW INTELLIGENCE

Cash-flow analysis complements accounting profitability by examining how cash is generated and used. The platform analyses operating, investing and financing cash flows and derives free cash flow.

15.1 Free Cash Flow

Free cash flow is represented as operating cash flow plus investing cash flow according to the project specification.

15.2 Capital Allocation

- Reinvestment.
- Shareholder returns.
- Debt repayment/financing changes.
- Potential distress patterns.

15.3 Cash Quality

The relationship between reported profit and cash generation is examined to identify situations that require further investigation.

16. TREND AND ADDITIONAL ANALYTICS

Historical trends help identify whether financial performance is improving, stable or deteriorating. The platform uses multi-year observations to calculate growth measures and present historical movement.

16.1 Trend Indicators

- Revenue.
- Profit.
- EPS.
- Margins.
- Debt.
- Cash flow.

16.2 Clustering

The project produces cluster assignments for the company universe. Clusters are exploratory groupings based on selected analytical characteristics.

16.3 Outlier Awareness

Extreme ratios and unusual changes should be investigated rather than interpreted automatically.

17. DASHBOARD DESIGN

The Streamlit dashboard provides interactive access to the analytical layer. It is organised around common financial-research tasks.

Screen	Purpose
Company Profile	Company-level overview
Screener	Filter companies
Peer Comparison	Peer benchmarking
Trends	Historical movement
Sectors	Sector analysis
Capital Allocation	Cash-flow behaviour
Annual Reports	Document access
Valuation	Valuation-oriented analysis

The dashboard reads the prepared database and analytical outputs rather than performing the entire ETL process during each user interaction. This improves response consistency and keeps data preparation separate from presentation.

18. API AND SOFTWARE INTERFACE

FastAPI provides a REST interface over selected project functionality. This makes the analytical platform usable by other applications in addition to the Streamlit dashboard.

18.1 API Features

- HTTP REST interface.
- JSON responses.
- OpenAPI documentation.
- Health endpoint.
- Company and analytical resources.
- Screener and ratio access where supported.

18.2 Validation

API tests verify health, valid resource retrieval and appropriate handling of invalid identifiers.

19. REPORTING AND EXPORT

Reporting converts analytical outputs into formats that can be reviewed, shared and archived.

19.1 Company Tearsheets

The project generates company-level PDF tearsheets for the 92-company universe.

19.2 Excel and CSV

Spreadsheet and CSV outputs support filtering, sorting, audit review and external analysis.

19.3 Portfolio Summary

The portfolio output provides a consolidated view of the company universe.

19.4 Document Repository

Annual-report and supporting-document metadata are exposed through the dashboard.

20. TESTING AND VALIDATION

Testing is essential because financial calculations can appear plausible even when formulas or data alignment are incorrect. The project includes automated tests covering ETL, KPI calculations, data quality and API behaviour.

Validation Item	Result
Company universe	92
Clean financial-ratio rows	1,070
Stock-price rows	5,520
Peer-group source rows	56
Company tearsheets	92
Radar charts	92
Automated tests	119 passed
Critical DQ violations	0

The clean ratio count is below the specification target of 1,100+ because the supplied data does not provide enough valid observations. The implementation does not fabricate missing records.

21. RESULTS AND OUTPUTS

The completed implementation provides a functioning end-to-end financial analytics application. The pipeline prepares the database, the analytical modules generate financial intelligence and the application layer makes the results accessible.

21.1 Major Outputs

- nifty100.db central database.
- ETL audit and validation outputs.
- Financial-ratio analytics.
- Screener results.
- Peer comparison outputs.
- Radar charts.
- Cash-flow intelligence.
- Company reports.
- Streamlit dashboard.
- FastAPI service.
- Automated test report.

21.2 Practical Outcome

The project demonstrates a complete workflow from raw financial spreadsheets to an interactive analytical application.

22. LIMITATIONS

The implementation follows the supplied data and specification and therefore inherits limitations present in the source material.

22.1 Identified Limitations

- Sector source contains 10 broad sectors while the specification describes 11.
- Some source records cannot be linked to the 92-company master.
- Clean ratio count is 1,070 rather than 1,100+ target.
- Some qualitative source information is incomplete.
- Supplementary market data are simulated rather than live.
- The system is an analytical aid and not investment advice.
- Power BI is handled separately.

22.2 Handling

The system records or excludes unsupported observations rather than fabricating values.

23. CONCLUSION

The Nifty 100 Financial Intelligence Platform brings together data engineering, relational database design, financial analysis, visualisation, reporting, API development and testing in one internship project.

The platform converts heterogeneous financial datasets into structured company, peer, sector and cash-flow intelligence. Its KPI engine provides repeatable calculations, while screening, health analysis, peer comparison, sector analytics and cash-flow modules provide complementary perspectives.

The project also demonstrates that reliable financial analytics depends on data quality and traceability in addition to calculations. Source limitations are explicitly documented instead of being hidden.

24. FUTURE SCOPE

The current platform provides a strong foundation for further development.

24.1 Enhancements

- Approved live/scheduled data providers.
- Automated periodic ETL refresh.
- Cloud deployment.
- Authentication and role-based access.
- Advanced alerts and notifications.
- Improved annual-report extraction.
- Expanded scenario and valuation modelling.
- Power BI integration.
- Enhanced audit trails.

25. REFERENCES

Internship Project Specification: Nifty 100 Financial Intelligence Platform.

Supplied Nifty 100 financial and supporting Excel datasets.

Python, pandas, NumPy, SQLite, Streamlit, FastAPI, scikit-learn, ReportLab and pytest documentation.

Project source code, configuration files and generated analytical outputs.

APPENDIX A – PROJECT DEMONSTRATION FLOW

The following sequence can be used during the project demonstration:

1. Activate the Python virtual environment.
2. Run the data pipeline when the database needs regeneration.
3. Start the Streamlit dashboard.
4. Open Company Profile and select a company.
5. Demonstrate financial KPIs and trend information.
6. Open Screener and apply a predefined screen.
7. Open Peer Comparison.
8. Open Trends and Sectors.
9. Open Capital Allocation.
10. Open Annual Reports.
11. Open Valuation.
12. Start FastAPI and open API documentation.
13. Run the automated test suite.
14. Show generated PDF, Excel and CSV outputs.

APPENDIX B – PROJECT FOLDER OVERVIEW

Folder / File	Purpose
data/	Raw and processed data
src/	Application source code
src/etl/	ETL and validation
src/analytics/	Financial calculations
src/dashboard/	Streamlit dashboard
src/api/	FastAPI application
src/reports/	Report generation
tests/	Automated tests
config/	Configuration
reports/	Generated reports
output/	Generated analytical outputs
nifty100.db	SQLite database
requirements.txt	Python dependencies
README.md	Project setup and usage